EXTENDING THE CONVERSATION AROUND THE ETHICAL USE OF AI IN ABA

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ABSTRACT

Artificial intelligence is increasingly used in the delivery and operational support of applied behavior analysis (ABA) services. The form of AI most practicing behavior analysts now encounter is large language models (LLMs) and the multimodal systems within which they are embedded. Existing publications on AI ethics in ABA have largely addressed AI broadly, and appropriately so for the stage of the conversation when written and scope of the documents. However, LLMs raise unique ethical considerations that benefit from direct analysis. This article takes that step to extend past work on AI ethics in ABA. We first describe how LLMs and multimodal systems work at a functional level. We then identify a set of ethical challenges these systems pose in ABA and analyze four challenges in depth: hallucinations and their interaction with clinical de-skilling and never-skilling; bias embedded in internet-scale training data; the opacity of LLMs and its implications for informed consent; and the specific risks of AI-generated clinical documentation. Each challenge is examined through converging lenses drawn from clinical ethics, behavior analytic ethics, and AI ethics. Finally, we provide a technology-specific, multi-framework analysis that organizational AI committees can use in their day-to-day operating practice. To aid in translation we offer a worked example of how such a committee might evaluate an emerging AI-driven tool.

Keywords artificial intelligence · AI governance · large language models · ethics

Artificial intelligence (AI) is no longer a novelty in applied behavior analysis (ABA). Scheduling algorithms seek to optimize therapist-client pairings, machine learning (ML) models seek to assist with data analysis, and a growing number of platforms promise AI-generated session notes, treatment recommendations, and clinical decision supports (e.g., Weil et al., 2025). For example, among behavioral health professionals attending the National Council for Mental Wellbeing webinars, reports of daily organizational AI use nearly doubled from 15% to 29% between 2024 and 2025 (National Council for Mental Wellbeing, 2025), and shadow AI use by healthcare employees may run much higher (Wolters Kluwer, 2026). By any measure, AI has arrived in ABA.

What has changed most dramatically since the first peer-reviewed publication on AI ethics in ABA (Jennings & Cox, 2024) is the type of AI that most behavior analysts likely use. The release of ChatGPT in Fall 2022 (OpenAI, 2022) marked a turning point. For the first time, anyone with an internet connection could interact with a large language model (LLM) capable of generating fluent, human-sounding text on nearly any topic, including clinical topics well outside the AI system's competence. Since then, LLM-based tools have become the dominant form of AI that most practicing behavior analysts knowingly encounter, and they are fundamentally different from earlier AI systems in ways that carry distinct ethical implications.

A small body of work has addressed AI ethics in ABA (Table 1). Jennings and Cox (2024) applied a principlist framework (Beauchamp & Childress, 2019) to AI broadly. Cox (2025) argued that the fundamental skill behavior analysts need to use any AI system ethically is to understand how mathematical models are built, and proposed a framework of AI literacy levels specific to behavior analytic roles. The Council of Autism Service Providers (CASP, 2025) published practice parameters covering the full AI adoption lifecycle. Its glossary defines LLMs and hallucinations, and its training section addresses how staff should report hallucinations during AI use. And, the AI

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Consortium for Applied Behavior Analysis (AIC-ABA, 2025) published eleven ethical principles to guide practitioner use of AI in ABA.

These documents represent important and meaningful first progress toward practically addressing AI ethics specific to ABA. However, each was appropriately scoped to address AI broadly. For example, Jennings and Cox (2024) did not reference LLMs at all as the article had been submitted for publication before the release of ChatGPT. Cox (2025) described model building generally without discussing the nuanced ethical considerations specific to LLMs and multimodal systems. CASP's (2025) practice parameters and AIC-ABA's (2025) guidelines both treat AI primarily as an umbrella category (and for good reason). CASP (2025) also engages directly with LLMs and hallucinations, however, the focus is largely operational rather than analytical. More specifically, whereas prior documents make suggestions for how practitioners can respond to issues that arise during AI use, this article asks why the underlying mechanisms produce ethical risk in the first place. We also ask how practitioners might reason through that risk when evaluating specific tools. The goal is more to provide a template process AI committees (see below) might use to analyze AI tools while highlighting the distinct ethical challenges raised by LLMs and multimodal AI systems.

AI type distinctions matter because LLMs and multimodal systems differ from other types of AI systems (e.g., discriminative AI; classic ML) in important ways (for reviews and discussion of the latter see Cox & Jennings, 2024; Jennings & Cox, 2024). For example, a scheduling algorithm that produces a suboptimal therapist-client match fails within a bounded and characterizable output space. In contrast, an LLM that generates a plausible-sounding, but fabricated, treatment rationale fails in a way that might be invisible to the practitioner relying on it. These are categorically different risk profiles and require ethical analysis specific to the technology producing them.

The purpose of this article is to provide one model of such an analysis. To get there, we first describe how LLMs and multimodal systems work at a functional level. We then identify a set of ethical challenges that LLMs uniquely pose in ABA and analyze four major challenges through the converging lenses of clinical ethics, behavior analytic ethics, AI ethics, and the normative ethical frameworks that underlie them. To help make this work tractable, we then describe how ABA organizations might establish AI committees and close by working an example of AI tool evaluation and how AI committees might use the previously described process to arrive at a system for their company. The hope is that this article helps readers begin moving from asking "is AI ethical?" to asking "what specific ethical challenges does a specific tool create, and how should we analyze and respond to them?"

How Large Language Models Work

The ethical challenges LLMs pose are difficult to fully understand without a basic understanding of how these systems function. This does not mean every behavior analyst needs to become an ML engineer. Rather, the level of understanding required to evaluate a tool's ethicality is likely proportional to the potential risks and harms of its use (Cox, 2025). The bar for understanding how AI systems work is much higher when the system's outputs influence treatment decisions, clinical documentation, or communication with families than when an AI tool suggests a restaurant for dinner. What follows is a brief overview of LLMs and multimodal systems to help ground the ethical analysis in subsequent sections. Readers seeking a more detailed treatment of mathematical model building and resources around related AI literacy in behavior analysis are referred to Cox (2025).

Large Language Models

At their core, LLMs are sophisticated pattern-matching systems (Figure 1). LLMs are trained on enormous datasets, often encompassing large portions of the publicly available internet including books, websites, academic papers, forums, social media posts, and code repositories (Vaswani et al., 2017). From this training, LLMs mathematically relate statistical relationships between words, phrases, and concepts[2]. When given a prompt, an LLM generates a response by predicting, one token at a time[3], what token is most likely to come next given the text that precedes it, while injecting variability into which high-probability token is selected so as to read less rote.

Token prediction is fundamentally different from retrieving information from a database. For example, when you search Google Scholar for journal articles relevant to a question you have, Google's database search function retrieves a set of records that exist in its stored data, curates that list based on your past browsing and clicking history, and then

[2]Note that nothing spooky is being referenced here. We are referencing this in the related behavior analytic conception of "concept" as perceptual concepts, relational concepts, or functional concepts (Zentall et al., 2002) as emitted textually such that we can group verbal relations in a form of mathematically-driven (but incomplete) type of concept learning (e.g., Layng, 2019; Tiemann & Markle, 1990).

[3]For the unfamiliar, tokens in LLM jargon are "words, character sets, or combinations of words and punctuation" (Microsoft, 2026). For a neat visualization of this process, see 3Blue1Brown (2024).

displays the final list in the browser window. In contrast, when you ask an LLM for a similar article citation, the model systematically generates text that looks like a citation based on the statistical textual patterns identified during model training. The LLM may produce a real citation, a partially correct one, or an entirely fabricated one. And it will present all three with identical fluency[4]. The LLM has no mechanism for distinguishing whether the output is true or fabricated because it is generating tokens based on next-token probability, not retrieval and regurgitation of content from existing articles in a database.

Subsequently, four properties of LLM architecture carry ethical weight. First, the training data determines the boundaries of the model's behavior. LLMs can only generate text that reflects statistical token relations present in the training data. If that data contains biases (and all datasets do), the model will reproduce those biases in its outputs (Bender et al., 2021). Second, LLM outputs are probabilistic, not deterministic. Two behavior analysts asking the same clinical question to an LLM at the exact same time may receive different responses. And the same behavior analyst asking the same question at two different points in time may receive different responses. This is because the generation process involves sampling from a probability distribution over possible next tokens. Lowering the sampling temperature can reduce variability, but hosted LLMs are often not fully deterministic even at a temperature of zero (Atıl et al., 2025; Omar et al., 2025; Ouyang et al., 2025), and many products do not let users adjust it (Davis et al., 2024; Srinivasan et al., 2026). Third, LLMs cannot independently verify the accuracy of their outputs. They generate statistically plausible text rather than text verified against validated sources of truth (though see below on agentic systems and attempts to address this). Fourth, LLM outputs appear authoritative regardless of accuracy. Hallucinated (i.e., factually incorrect; but see Emsley, 2023, on why "fabrication" might be the more accurate term) content reads with the same fluency as correct content, and qualifying autoclitics (e.g., hedging cues) are largely absent unless the model has been specifically trained or prompted to produce them.

As a final note, LLMs as described above are text-based. Increasingly, however, the AI tools that practitioners encounter are likely to be multimodal. That is, the AI system can process images, audio, and video alongside text (OpenAI, 2024). For example, an AI that analyzes video of a therapy session might code behavior or flag potential concerns. An AI that processes audio might transcribe and summarize sessions in real time. And, an AI that processes images might analyze data sheets or video-recorded environmental arrangements. For ABA specifically, multimodal capabilities create additional ethical concerns similarly extending from the probabilistic, generative nature of how these AI systems operate. The challenges discussed in the following section apply to both text-based LLMs and multimodal systems. Where multimodal capabilities create additional or distinct concerns, they are noted.

It is also worth noting that few practitioners interact with a base LLM directly. Most encounter LLMs through deployed systems that wrap the model in additional components and where those components change the functional properties of the system. For example, retrieval-based augmented generation supplies the model with documents from a curated source (e.g., an organization's policies, a published guideline, a client's own records) before it generates a response (Lewis et al., 2020). Other systems add citations to the sources used, structured output templates, checks that flag statements not supported by the retrieved sources, the option for the model to decline to answer, audit logs, and required human review before an output is used.

These safeguards do not eliminate risk. They can, however, change how often errors occur, where they occur, and how easily they can be detected. Thus, hallucination and opacity are better thought of as varying along a continuum with system design and workflow than as all-or-non properties of LLMs. Table 2 describes five types of LLM-based systems that behavior analysts are likely to encounter, the safeguards each might include, and their relative risk for clinical or educational use. Throughout the remainder of this article, we discuss properties of the underlying models when referring to LLMs generally and note where specific deployment features change this analysis.

Understanding how LLMs and multimodal systems function is a technical exercise that creates the starting point for ethical analysis. Ethically evaluating AI tools requires a sequence of analytic steps. First, one must understand the functional properties of the system (i.e., what inputs it receives, how outputs are generated, and how information is stored or updated over time). Second, those properties are examined to identify the specific ethical risks that arise from them. Third, those risks are analyzed through converging ethical frameworks such as behavior analytic ethics, clinical ethics, and AI ethics. Finally, that analysis should be translated into governance decisions about whether and how the technology should be used. The sections that follow illustrate this function-risk-governance analytic process by examining several ethical challenges created by LLM-based tools. Later in the article, we return to this framework in a worked example demonstrating how an organizational AI committee might apply the same analytic method when evaluating a specific AI system.

[4]In AI jargon, fluency is defined slightly differently than in behavior analysis. In behavior analysis, fluency generally is defined relative to response accuracy, speed, and latency (e.g., Binder, 1993; Johnson & Layng, 1996). In contrast, fluency in AI generally refers to AI system output speed and the absence of qualifying autoclitics (e.g., hedging, uncertainty, masking hallucinations) (Anthropic, 2024; Salvi et al., 2025).

Ethical Frameworks Used in This Analysis

Claims of ethicality throughout this article necessarily turn on the normative justification for the claim being made (Cox, 2021). Thus, before turning to specific challenges, it is worth being explicit about the frameworks we use and how they relate. From a functional-contextual perspective, normative ethical theories can be considered as verbal rules that specify which features of a situation should control behavior and which outcomes make a response "right". Behavior analysts do not have to commit to a single theory to use normative ethical theories this way. Rather, each framework offers a different set of questions a practitioner or committee can ask of the same AI tool.

Principlism is the dominant framework in Western clinical ethics and organizes ethical analysis around four principles (Beauchamp & Childress, 2019). Beneficence asks whether an action benefits the client. Non-maleficence asks whether an action avoids causing harm. Respect for autonomy asks whether an action supports the informed, self-determined choices of clients and caregivers. And, justice asks whether benefits, risks, and costs are distributed fairly. Jennings and Cox (2024) used principlism to frame AI ethics in ABA and is used here for continuity. The AI ethics literature has adapted principlism by adding a fifth principle of explicability (e.g., Floridi & Cowls, 2019). Explicability combines intelligibility (i.e., can those affected understand how the system works?) with accountability (i.e., who is response for how it is used?).

Principlism is a mid-level framework. Its principles are often justified by appeal to broader normative theories which justify an action using different benchmarks. From a utilitarian perspective, justification turns on the relative ranking of benefits and harms resulting from one's behavior. From a deontological perspective, justification turns on compliance with the duties that apply given the contextual circumstances (e.g., duty to obtain informed consent), regardless of the consequences in a particular case. From a virtue theory perspective, justification turns on the emission of specifically identified virtuous behaviors. Practically, this means asking which repertoires (e.g., honesty, attentiveness, practical judgment) should be acquired and maintained (Anscombe, 1958). And, from an ethics of care perspective, justification turns on promoting the well-being of care-givers and care-receivers in a network of social relations (Noddings, 1984).

We use these frameworks as converging lenses. When a benefit or a concern is identified from several frameworks that justify ethical claims in different ways, the conclusion is arguably more robust as different theories making different assumptions lead to the same claim of "right" or "wrong". Convergence is not guaranteed, however. Frameworks and principles can make differing claims on what is "right" in the same context (e.g., Cox, 2021). For example, an AI documentation tool might advance beneficence by giving clinicians more time with clients while undermining autonomy if families are not told how their data are used.

As noted by Kelly et al. (2021), applying ethical rules without guiding principles can result in an overly simplified and mechanistic approach to ethical problem solving. Those authors also described how one ABA organization developed explicit principles to guide employee ethical decision making. AI committees can similarly make explicit how they resolve tensions between ethical frameworks and suggest three practices when ethical frameworks diverge. First, name the incompatible behaviors explicitly rather than letting one framework win by default. Second, look for safeguards or modifications that reduce the conflict (e.g., an opt-out that preserves autonomy while retaining most of the benefit). Third, document the reasoning so the decision can be revisited as the tool, the evidence, or the organization changes.

Ethical Challenges and Analysis

The ethical challenges posed by LLMs and LLM-driven tools in ABA are numerous and are increasingly discussed broadly across the clinical ethics and AI ethics literatures (e.g., Armitage, 2025; Fareed et al., 2025; Haltaufderheide & Ranisch, 2024; Ong et al., 2024; Weiner et al., 2025). Commonly discussed current topics include: hallucination and factual unreliability (e.g., Omar et al., 2026); automation bias and clinical de-skilling (e.g., Qazi et al., 2026; Rosbach et al., 2025); bias embedded in training data and amplified through outputs (e.g., Guo et al., 2026; Weidinger et al., 2021); the opacity of model decision-making and its implications for informed consent (e.g., Ghassemi et al., 2021; London, 2019); privacy and data security when clinical information is transmitted to third-party AI systems versus maintained on medical devices (e.g., Muehlematter et al., 2021; Weiner et al., 2025); the absence of clear accountability frameworks when AI contributes to clinical errors (e.g., Jennings & Cox, 2024; Raji et al., 2020); the use of AI in prior authorization and insurance processes that directly affect access to clinical services (e.g., Lubell, 2025; Raza et al., 2026); the risks posed by emerging agentic AI systems that operate autonomously over extended interactions (e.g., Collaco et al., 2026; Hinostroza Fuentes et al., 2025); the potential for AI companions to form harmful relationships with vulnerable populations (e.g., Malfacini, 2025; Ramsey, 2025); and the environmental costs of large-scale AI deployment (e.g., Liu & Yin, 2024; Ren et al., 2024).

Table 3 summarizes the above primary challenges. Each is categorized by the type of ethical concern and by whether the challenge is unique to or heightened by LLMs. Not all challenges can be analyzed within a single article. Thus,

what follows is an integrated analysis of four challenges selected because they are clearly specific to or substantially heightened by LLM-based tools; and they are clinically concrete, analytically rich, and collectively representative of different types of risk. The goal is not to be exhaustive. The goal is to model one method of analysis that draws on clinical ethics, behavior analytic guidelines, and the AI ethics literatures through converging ethical frameworks. As a whole, it provides one method whereby practitioners and organizations might analyze AI-tool-related challenges as they arise.

Hallucination and the Erosion of Currently Important Clinical Skills

Hallucinations might be the most immediately dangerous LLM concern in clinical settings as well as the most difficult one to detect. As noted above, hallucinations are a structural feature of how LLMs generate text, not an occasional glitch. Documented rates of hallucination in healthcare vary widely by task and testing strategy, ranging from roughly 1.5% in structured clinical note generation (Asgari et al., 2025) to upwards of 40% on general clinical tasks and over 80% under adversarial conditions (Omar et al., 2025). Fabricated citations are a particularly well-documented problem. For example, Walters and Wilder (2023) found that GPT-3.5 fabricated 55% of bibliographic citations and GPT-4 fabricated 18%. In mental health literature specifically, GPT-4o fabricated roughly one in five academic citations, and 55% of generated citations were either fabricated or contained errors (Linardon et al., 2025). In agentic systems with persistent memory, these errors can also compound over time (see the worked example below).

Clinical risk via LLM use is created from the asymmetry between the appearance of competence and the reality of unreliability. Hallucinated treatment recommendations read identically to empirically supported ones. This asymmetry interacts with the human tendency to over-rely on technology system outputs and to accept them uncritically (i.e., automation bias; Goddard et al., 2014). For example, pathologists have been shown to abandon initially correct decisions when confronted with erroneous AI suggestions (Qazi et al., 2026). And, a 2025 mixed-method review concluded that de-skilling from AI reliance is already being detected in medicine (Natali et al., 2025). For ABA, this concern is heightened by the fluency of LLM outputs. A three-paragraph, well-structured treatment rationale with apparent clinical reasoning and real looking citations reads more like the work of a competent colleague than a traditional decision-support alert. The more human the output reads, the lower the likelihood that the errors are caught (e.g., Cohn et al., 2024).

Concerns around automation bias and de-skilling further extend to never-skilling. That is, less skilled practitioners (e.g., behavior analysts in training, new behavior technicians) never developing certain skills previously required for effective practice because AI systems automate that skill practice and performance. Never-skilling interacts with hallucinations in important ways. For example, a BCBA with fifteen years of independent practice can draw on that experience when evaluating an AI system output. In contrast, a newly certified BCBA who has used LLM tools throughout graduate training and their early career may never develop the independent clinical reasoning that would allow them to catch the same errors. Vallor (2015) anticipated this dynamic as moral deskilling, the atrophy of character traits needed for ethical decision-making when technology mediates professional judgment. Saadeh et al. (2025) drew a related distinction between decision support and decision substitution. When an algorithm is sufficiently complex that no clinician can assess all factors within a reasonable timeframe, the result might be substitution of professional judgment rather than support of it.

Never-skilling is, at its core, a supervision challenge. Supervised fieldwork is where trainees develop the independent clinical reasoning that later allows them to catch errors in AI outputs (Sellers et al., 2016). If trainees complete case conceptualization, treatment plans, and progress summaries with LLM assistance from the start, the supervisory contingencies that shape independent reasoning may never contact the trainee's own behavior. Supervisors are, thus, well positioned (and, in some instances, arguably obligated) to arrange deliberate AI-free practice. For example, a supervisor might require a trainee to draft a case conceptualization independently before comparing it with an AI-generated version and then explain any differences. And, organizations and graduate programs might similarly specify which fieldwork activities must be completed without AI assistance and which skills developed to fluency before AI use is allowed.

We can integrate the three ethical lenses of applied behavior analytic ethics, clinical ethics, and AI ethics relative to hallucinations and the erosion of current clinically important skills. Within behavior analytic writings, the Behavior Analyst Certification Board (BACB, 2020) Ethics Code requires behavior analysts to practice within their competence and base clinical decisions on the best available evidence. This does not mean behavior analysts have to reconstruct every tool they use. No BCBA independently derives the norms of a standardized assessment or checks the numerical methods inside their graphing software. Rather, behavior analysts use these tools competently because their trust in them is warranted. Standardized assessments and statistical software have been externally validated; many have undergone regulatory or peer review; and their errors are known, within an acceptable margin of error, and relatively stable.

LLM-based tools in ABA currently have none of these features. Their clinical outputs have not been externally validated, they change with vendor updates, and their errors are neither stable nor easily characterized. Thus, the competence problem is a result of the infrastructure not yet existing that would warrant trusting them (see Weil & Cox, 2026, for one approach to evaluating clinical decision models). Until that infrastructure exists, if a behavior analyst cannot independently evaluate an AI-generated recommendation, acting on that recommendation arguably falls outside their competence. And, consistent with the proportionality principle described above (Cox, 2025), this obligation increases as the clinical consequences of the decision increase. CASP's (2025) training guidance on reporting hallucinations, described above, provides an important structural check.

Applying a principlist lens (Beauchamp & Childress, 2019), hallucinations implicate all four foundational clinical ethics principles at once (see also Kozlakidis et al., 2026). Beneficence and non-maleficence are implicated through harm from inaccurate recommendations. Autonomy is implicated through the consent implications relative to unreliable or inaccurate content. And, justice is implicated because hallucination rates are not uniform across topics or populations (via bias in training data), and errors are more frequent and harder to detect for populations less well represented in the training data (e.g., Omar et al., 2026; Weidinger et al., 2021).

More broadly in AI ethics, Floridi and Cowls (2019) identified explicability as the most distinct AI ethics principle. Here, explicability combines intelligibility with accountability. Hallucination directly undermines explicability. If neither the model nor the practitioner can reliably distinguish accurate from fabricated content, the system fails the most basic test of being intelligible to those who use it. Practitioners behave ethically when they are responsive, judge behavior accurately in context, and apply a functional-contextual understanding of behavior-environment relations. These are exactly the behavior patterns at risk of being displaced when clinical reasoning is outsourced to a tool that probabilistically generates generic, pattern-based next-token responses.

Importantly, the practical question is not whether hallucination disqualifies LLMs from clinical use. LLMs offer many potential benefits to ABA service delivery such as improving the efficiency, detail, and accuracy of documentation in mental healthcare settings (e.g., Hua et al., 2025); improving behavioral healthcare service delivery (e.g., Stade et al., 2024); and clinical decision support (e.g., Goh et al., 2025). These benefits each carry ethical weight. Human documentation and clinical reasoning are also vulnerable to fatigue, inconsistency, delays, and competing contingencies. A fair ethical analysis, therefore, has to compare AI-assisted practice with practice as it currently occurs rather than with idealized clinical judgment. That is, it asks when AI reduces existing risks, when it introduces new ones, and when it simply shifts risk from one place to another.

The practical question instead is how to use LLMs while keeping independent clinical judgment as the authoritative source of decisions. At minimum, this requires that practitioners understand that hallucinations are structural features of generative AI systems as opposed to occasional glitches. Subsequently, AI-generated content must be verified against independent sources before using AI system outputs to act. In turn, organizations need to build contingencies supportive of AI system output verification into their employees' workflows rather than relying on individual vigilance and potentially incompatible contingencies surrounding everyday clinical decision-making (Table 4).

Bias in Training Data and Systems

All AI systems are shaped by the data used to build them. For classical ML models in ABA, the training data is typically a curated clinical dataset and its limitations are knowable. For example, consider an AI system built to recommend hours of ABA or treatment goals using data from a clinic serving 2-to-6 year-olds from primarily white, suburban, English-speaking families. The training dataset used to build this AI system can be transparently described, and caution can be advised when generalizing outside that client-defined and setting-defined scope. The scope of AI system bias is bounded by the scope of the data.

LLMs are different. LLM training data is, by-and-large, the internet along with as many books, academic publications, and other text corpora that large AI companies can ingest. The biases present in this textual data are not neatly bounded and explicitly scoped and defined. Rather, the biases that exist stem from the entire corpus of digitized human text: historical inequities, cultural assumptions, stereotypes, and representational imbalances accumulated across centuries (Bender et al., 2021) as well as the general absence of editorial accuracy comprising much published on the internet (e.g., social media forums, blogs). When an LLM generates text about autism, behavior, families, or clinical practice, it does so through the lens of the most common patterns present on the internet and related materials. And, because tokens are generated probabilistically, more recent, modern understanding of any topic might be less well represented probabilistically and, thus, be less likely to be displayed to the LLM user.

Importantly, researchers have confirmed that these concerns are not theoretical. For example, Brandsen et al. (2024) found high levels of negative bias against neurodivergent terms across 11 language model encoders. Concerningly, sentences such as "I have autism" or "I have epilepsy" showed stronger negative associations than "I am a bank

robber", with bias persisting even for commonly assumed virtuous behaviors such as "honesty" if described within the context of a neurodivergent behavioral pattern. As a result, LLMs that are used to generate documentation or caregiver communications are likely to embed these biases and produce subtle stigmatizing language that a quick review may not catch. Existing diagnostic disparities compound the problem. White children are 65% more likely than Hispanic children and 19% more likely than Black children to receive autism diagnoses (Durkin et al., 2017). Thus, even when clinically accurate, peer-reviewed information is used in the training data, disparities around underrepresentation of certain populations and the resulting clinical biases are encoded in the data from which LLMs learn statistical patterns to generate their outputs.

The behavior analytic AI ethics literature has commented on AI bias at a general level. For example, Jennings and Cox (2024) discussed model bias and generalizability as a key ethical concern. Cox (2025) argued that understanding model building helps practitioners identify sources of bias. CASP (2025) addresses equity and bias in the context of AI system selection and monitoring. And, AIC-ABA (2025) names equity and bias mitigation as one of its core principles. None of these sources, however, addresses the specific character of LLM bias: internet-scale, culturally embedded, often invisible in the fluency of the output, resistant to simple technical fixes, and significant difficulty in knowing and defining the scope of bias in outputs. When evaluating a classical ML tool, an organization can ask what data the model was trained on. For LLMs, these questions are often unanswerable because internet-scale training data is too large to characterize precisely.

From a principlist perspective, bias in LLMs implicates justice most directly. Systematically different quality of care for different populations is an inequitable distribution of benefit and harm. LLM output bias also implicates non-maleficence. Subtly stigmatizing language in AI-generated documentation or inappropriate clinical recommendations count as harm, even if unintended, and even if no individual practitioner set out to produce it. From an ethics of care perspective, bias is a failure of attunement. Generic, pattern-based responses may not reflect the particular needs, values, and context of the individual client and family which, in turn, reduces the overall improvement in well-being from receiving services (e.g., Constantino & Charman, 2016). As stated directly by Floridi and Cowls (2019), AI systems should not create or exacerbate unfair discrimination. LLMs make identifying and correcting potential output bias more difficult than deterministic AI systems (e.g., classical ML).

For organizations evaluating LLM-based tools, bias assessment is more than a one-time checkbox at the point of purchase. Bias assessment is an individual-level clinical skill that must be taught and ongoingly emitted by all practicing behavior analysts because LLMs are updated, fine-tuned, and modified by their developers, often without notice to end users, in ways that may change the character and severity of their biases. Topographically, this repertoire might include comparing AI-generated language against the raw data and against the family's stated preferences (e.g., identity-first versus person-first language), flagging deficit-based or stigmatizing descriptions, checking whether recommendations assume cultural norms the family does not share, and noticing when outputs for some clients are consistently less specific or less accurate than for others. And, to take action on common patterns, organizations need processes to monitor AI outputs across the populations they serve, flag patterns of concern, maintain human clinical judgment that can catch what automated systems miss, and efficient systems to rollback use of AI systems when the amount of bias crosses a threshold (Table 4).

Informed Consent and the Opacity of LLMs

Informed consent requires that clients or their surrogates receive sufficient information about a proposed intervention to make a meaningful decision. The "reasonable person standard" has long guided consent practices in healthcare (Odwazny & Berkman, 2017). But what happens when the tool being used is one that neither the practitioner nor its developers can fully explain? This is the consent challenge that LLMs create.

The opacity of LLMs operates at multiple levels. Models with hundreds of billions of parameters are fundamentally difficult to interpret even for the engineers who build them. To be fair, researchers are actively working on methods to make LLM architectures more explainable as to how the output is generated (e.g., transformer architectures, deep neural networks; Doshi-Velez & Kim, 2017; Lipton, 2018). However, current model explanation techniques often fail to produce explanations reflecting underlying model behavior (Ghassemi et al., 2021) or to explicitly know what patterns in textual data the model has learned (London, 2019). Further, even where model explainability techniques show promise, most behavior analysts using LLM-based tools do not have access to the model's underlying architecture or training data and, thus, would not be able to interpret that information if analysis and monitoring of that information is not made available by AI system vendors.

The above subsequently creates a layered consent problem. The practitioner cannot fully explain the tool because the practitioner does not fully understand it. The developers cannot fully explain the tool's behavior because its internal weights and learned textual relations are likely opaque even to them without setting up robust perturbation analyses.

And, even where model explainability methods are closing in on useful diagnostics, this information and techniques are not broadly available or embedded in technologies to be useful at the moment of clinical decision-making. Overall, the client or caregiver is, therefore, being asked to consent to care that is influenced by a process no one in the room completely understands or can explain as to why the care being recommended or delivered is most ethically justifiable. Without understanding how the LLM output is being derived relative to the clinical context, it becomes difficult to know whether an alternative course of action would have better met the normative ethical benchmark being set.

As with before, we can lean into various ethical lenses to analyze this challenge. From the behavior analytic ethical lens, the BACB Ethics Code (2020) requires behavior analysts to describe assessments and interventions prior to implementation and obtain informed consent before proceeding. Jennings and Cox (2024) raised the consent question directly, asking how one might describe the "intent" of an AI system sufficiently to warrant consent as informed. CASP (2025) and AIC-ABA (2025) each name transparency and explainability as core principles of AI ethics in ABA. From an AI ethics lens, Floridi and Cowls (2019) identified explicability as the principle most distinctive to AI ethics. LLMs fail the intelligibility test in a way previous clinical technologies did not because their outputs cannot be traced to specific inputs or decision rules.

From an ethics of care perspective comes greater concern. As past authors have noted, informed consent in ABA is not a one-off, procedural requirement (Graber & Maguire, 2024). From the ethics of care perspective, informed consent is part of the foundational relationship that ethical clinical practice rests upon. The conversation between a practicing behavior analyst and a family about what will happen in treatment, why, and what to expect is itself clinical behavior. Those ongoing conversations build trust, establish shared understanding, and respect the family's role as partners in care. When AI mediates that process in ways the practitioner cannot explain, the relational quality of the consent interaction diminishes, even if the legal requirements are technically met.

Finally, many ABA clients are minors or individuals with limited verbal repertoires. Surrogate decision-makers consent on behalf of individuals who may not be able to evaluate AI-related disclosures themselves. No one currently has a complete answer to this quagmire, given the long-term impact that sharing such sensitive client data may have. But, based on the converging literature, AI-related consent disclosures in ABA should at minimum cover: what AI tools are used and for what purposes; what client data, if any, is processed by or transmitted to the AI system; whether those data are retained or used to train the vendor's models; the continuing primacy of human clinical judgment in all decisions; known limitations of the tool, including the possibility of errors; how errors in AI-generated content are identified and corrected; the client's or caregiver's right to request that AI not be used in their care; and the privacy safeguards in place (Table 4).

Meaningful consent of this kind does not require that practitioners explain how the model works internally, which no one can currently do. Rather, meaningful consent requires that clients and caregivers understand what the tool does, what it does not do, and what protections are in place. Other healthcare disciplines are actively debating what informed consent requires when AI contributes to care (e.g., Cohen, 2020). And, at least one state now requires that patients be told when generative AI produces clinical communications that a licensed provider has not reviewed (Assemb. B. 3030, 2024).

Clinical Documentation and LLMs

If there is a single use case driving LLM adoption among practicing behavior analysts, it is likely clinical documentation. The appeal is obvious. Session notes, treatment plans, progress reports, and caregiver communications are essential but time-consuming. And, as has been well-documented in the basic choice literature, reducing the effort to contact reinforcement significantly increases response allocation to the reduced effort option (e.g., Hursh & Silberberg, 2008; Peck & Madden, 2024). Several AI platforms now market documentation capabilities specifically to ABA practitioners, and uptake appears to be growing rapidly.

It may help to distinguish the functions these tools perform as the ethical risks differ across them. Increasingly more clinical judgment is handed over to a system when moving across drafting (i.e., turning a clinician's notes or data into prose), summarizing (i.e., condensing existing records), coding (i.e., assigning data to categories such as behavior or prompt type), interpreting (i.e., describing what the data mean clinically), recommending (i.e., suggesting changes to treatment), and autonomous decision support (i.e., triggering clinical changes without a clinician's review). Under some conditions, AI drafting may improve the timeliness and consistency of notes and free clinician time for more meaningful review or clinical activities. The concerns below apply most strongly to the later functions in this list and to drafting that replaces, rather than supports, a clinician's review of the data.

The ethical considerations are not abstract. The hallucination problem is particularly acute for documentation. ABA notes require highly specific technical language, including information about reinforcement schedules, prompt hierarchies, operational definitions, and trial-by-trial data. Generic LLMs are unlikely to render this information accurately

because they are not physically present in session to observe the interactions between technicians and clients. An AI-generated note that describes a "token economy" when the actual procedure was differential reinforcement, or that reports "3 instances of aggression" when the raw data show 3 instances of non-compliance, is inaccurate and a potential legal liability. And, the BCBA is responsible for the integrity of all documentation regardless of how it was generated (BACB, 2020).

Bias is also relevant. There are several highly probable examples that seemingly need to be reviewed for every client and every note created. These include: bias-related characteristics such as an LLM trained on text that frames autism through a deficit lens that may subsequently generate documentation emphasizing limitations over strengths; using person-first language when the family prefers identity-first language (or vice versa); and framing caregiver concerns through cultural assumptions embedded in the training data rather than the family's actual perspective. These effects are easy to miss in a quick review, can accumulate across months of documentation increasing the likelihood they are used even more in future notes, and are exacerbated by automation bias that low-effort response contingencies are likely to reinforce.

Documentation also raises the de-skilling concern. Writing a session note is more than an administrative checkbox to meet billing requirements. Clinically, its functional purpose is arguably to provide a high-level summary of what took place in session as well as information for the supervising BCBA that may not be captured by the data collection systems currently used during sessions. In turn, this allows the supervising BCBA to review data, synthesize and identify patterns of behavior-environment relations at larger timescales, connect this information to the treatment plan, and make clinical judgments on how to adjust service delivery. This process of reviewing and synthesizing is itself a form of clinical reasoning. When organizational contingencies (e.g., productivity targets, billing structures, training that promotes AI-generated notes) leads technicians to outsource note writing to systems physically unable to capture and provide all relevant information, the supervising BCBA's clinical decisions are less well-informed. And, BCBAs who routinely outsource clinical reasoning to an AI tool may, over time, become less engaged with the details of their clients' progress.

From a principlist perspective, generative AI producing clinical documentation implicates all four principles. Beneficence asks whether the tool actually improves care or simply reduces effort. Non-maleficence asks what harms follow from errors that may go uncaught. Autonomy asks whether clients know their documentation is AI-generated. Justice asks whether errors and biases distribute equitably across the populations served. From an ethics of care perspective, documentation in ABA is a form of clinical attention. When a BCBA writes a session note, they review the data, notice what happened today that was different from yesterday, and let that comparison shape tomorrow's session. The writing is how the clinical relationship is sustained across visits. A practitioner who outsources the writing to an AI tool without reviewing the underlying data outsources the attention that made the writing clinically meaningful. The documentation may still be adequate. The relational work the documentation used to do is gone.

As a final important point, as of this writing, no peer-reviewed validation studies exist for the clinical documentation features of any ABA-specific AI tool, despite active marketing and use. This contrasts with medicine, where validation studies of AI-generated clinical documentation are beginning to appear (e.g., Asgari et al., 2025). A recent review found that evidence for AI-driven clinical decision support in ABA is similarly lacking (Weil et al., 2025). Organizations adopting these tools are conducting uncontrolled experiments on their own clinical documentation with the potential ethical and legal concerns noted above. Again, this does not mean that it is inherently unethical to use AI-generated clinical documentation tools. Rather, the absence of empirical validation should be made explicit in any consent conversation with clients and any decision-making process within the organization as well as how clinicians are validating that "correct" outputs are, in fact, correct (Table 4).

Establishing AI Committees

The ethical challenges described above are not problems to be solved once and filed away. LLMs are updated regularly, often without notice to users, in ways that can change their capabilities, limitations, biases, and risk profiles. New tools also enter the market continuously under the guise of being better or cheaper than existing models. Given that basic contingencies influencing choice also influence technology vendor behavior, it is likely the underlying models used will subsequently change to balance performance-cost tradeoffs with or without transparent communication with the technology product end-users. Ethical analysis of AI in ABA is, thus, an ongoing organizational function. The relevant practical questions become where that function lives and who sets the standards.

Previous authors have described the rationale for and structure of organizational ethics committees and related functions in behavioral health settings (Brodhead & Higbee, 2012; Cox, 2020; LeBlanc et al., 2020; Valentino et al., 2025). Readers seeking detailed guidance on committee infrastructure, membership, and implementation are referred to that work. CASP (2025) also provides complementary organizational guidance specific to AI, including a lifecycle

framework for selection, deployment, monitoring, auditing, error reporting, and deprecation. And, AIC-ABA (2025) offers a set of eleven principles that can inform committee deliberations. Between them, these three documents supply examples of committee structure (Brodhead & Higbee, 2012; Cox, 2020; LeBlanc et al., 2020; Valentino et al., 2025), an AI lifecycle framework (CASP, 2025), and a set of ethical principles (AIC-ABA, 2025). Still remaining to be made explicit in published AI-ABA ethics work is an analytical method to evaluate specific tools as they arrive. Below, we hope to help jumpstart that conversation as an example for how an AI-ABA committee might go about its day-to-day operations.

Why AI-Specific Committees?

One initial and fair question is whether a general ethics committee could simply add AI to its portfolio. In many organizations, that might be a practical starting point. But three considerations favor a dedicated or substantially differentiated approach. First, the pace of AI change (daily, weekly) outstrips the pace at which ethics committees typically operate (monthly, quarterly). Model updates can change ethical implications overnight, and committees need faster response cycles than weeks-long deliberations can support. Second, the technical knowledge required to evaluate AI tools is specialized. Committee members do not need to be AI experts, but the committee needs access to AI-relevant expertise through membership, consultation, or ongoing education. Third, AI introduces questions that cut across traditional committee functions. AI tools are simultaneously a clinical tool, a data system, an organizational investment, and an employee tool. Cross-cutting issues may fall between the cracks of existing committee structures or get addressed piecemeal by different departments without the integrated analysis the topic demands.

Function(s) of AI Committees

AI committees function as organizational mechanisms for systematically evaluating how emerging technologies interact with clinical practice, ethical obligations, and organizational policy. In contrast to one-time policy creation or vendor vetting, their primary function is the continuous analysis of data related to evolving technological systems within the context of clinical service delivery. This is necessary because AI tools—particularly LLM-based systems—change over time through updates, retraining, or vendor-level infrastructure changes that may alter AI system capabilities and risks (Bommasani et al., 2021; Floridi et al., 2018).

Drawing on the broader ethics committee literature in behavioral health and medicine, ethics committees in health service organizations have traditionally served several functions, including ethics education, policy development, case consultation, discussion forums for ethical concerns, and research oversight (Moon et al., 2019). Organizational ethics committees have been widely adopted in healthcare settings precisely because ethical rules are written broadly but must be interpreted in specific contexts encountered by practitioners (Cox, 2020). AI committees inherit many of these functions but apply them to a distinct technological domain in which tools simultaneously influence clinical reasoning, documentation, operational workflows, and data governance.

Within ABA organizations, the functions of an AI committee might be conceptualized along three interacting levels. First, AI committees serve a clinical ethics function. Here, the committee evaluates how specific AI tools may influence the welfare of clients, families, and practitioners. This includes identifying risks such as hallucinated recommendations, bias in generated language, or automation bias in decision making. Research across healthcare settings has demonstrated that clinicians may, under certain conditions, over-rely on AI outputs even when those outputs are incorrect (Goddard et al., 2014; Lebovitz et al., 2022). The committee's role is to anticipate how the introduction of a tool may alter behavioral contingencies surrounding clinical decisions. In this sense, AI committees function as a structured environment in which organizations can analyze how technological stimuli may alter practitioner behavior.

Second, AI committees serve a governance function. Organizations require clear policies regarding what tools can be used, what data can be entered into those tools, and how AI outputs must be verified before influencing clinical decisions. Governance frameworks for responsible AI adoption emphasize the importance of organizational structures capable of translating ethical principles into operational policies (Floridi et al., 2018; Jobin et al., 2019). The AI committee, therefore, operates as a bridge between emerging technologies and institutional policy development where committee members translate ethical analysis into practical guidance that shapes everyday clinical practice.

Third, AI committees serve a learning and adaptation function. LLMs are probabilistic AI systems whose performance characteristics are multidimensional and change over time. Consequently, organizations must monitor their real-world outputs and adjust policies or workflows accordingly. Scholars of sociotechnical systems have emphasized that responsible AI use requires continuous monitoring, incident reporting, and iterative adjustment as systems evolve (Amershi et al., 2019; Raji et al., 2020). AI committees provide a forum where organizations can accumulate experience with these monitoring, incident and error reporting, and iterative adjustment systems to refine organizational and clinical practices in response to new information and new AI systems.

Taken together, the three functions of *clinical ethics*, *governance*, and *learning and adaptation* position AI committees as organizational learning systems. Their purpose is both to approve or reject technologies and to continually examine how those technologies interact with clinical behavior, ethical obligations, and institutional structures.

Structure of AI Committees

The structure of an AI committee should balance two competing goals: access to sufficient expertise to evaluate complex technologies and representation of the organizational stakeholders affected by those technologies. Lessons from existing ethics committees in behavioral health suggest that committees function most effectively when membership reflects the range of perspectives within the organization rather than concentrating decision-making authority in a single individual (Cox, 2020; Moon et al., 2019).

At minimum, an AI committee typically requires four categories of participation. First, the committee should include clinical representation, typically one or more BCBAs or clinical directors responsible for treatment oversight. Because many AI tools directly influence documentation, treatment recommendations, or clinical decision support, individuals with expert knowledge of ABA practice are essential for evaluating how these tools interact with existing treatment models and professional ethical obligations (BACB, 2020).

Second, the committee should include technical expertise. This may take the form of a data scientist, AI engineer, or external consultant with experience evaluating machine learning systems. Committee members do not need to build AI systems themselves, but they must possess sufficient technical literacy to interpret vendor claims, identify common failure modes, and ask informed questions about model behavior (Amershi et al., 2019; Bommasani et al., 2021). The technical expert also plays an important role in creating sufficient AI system monitoring tools to track AI system performance over time.

Third, the committee should include organizational or operational leadership. Individuals here might be administrators responsible for compliance, billing, or operations. AI tools frequently influence documentation practices, reimbursement workflows, and data governance processes. Administrative representation ensures that analysis considers practical constraints within the organization.

Fourth, organizations may benefit from interest-holder (i.e., a more inclusive alternative to "stakeholder") representation such as individuals who represent the perspectives of technicians, caregivers, and clients. Interest-holder representation helps ensure that AI adoption decisions reflect the experiences of those directly affected by the technology rather than only those responsible for evaluating it. Inclusive representation is a common design feature of institutional ethics committees because it reduces the likelihood that ethical judgments reflect the learning history or contingencies of a single professional role (Cox, 2020; Moon et al., 2019).

Within this structure, a chair or coordinator typically organizes meetings, facilitates discussions, and coordinates follow-up activities. As with other ethics committees, access to legal or regulatory consultation might be useful when questions arise concerning privacy law, consent requirements, or data governance obligations (Moon et al., 2019). Importantly, the goal of committee structure is not to create a permanent group of technical experts. Instead, the committee functions as a cross-disciplinary deliberative body capable of integrating clinical, technical, and ethical perspectives into a coherent and ongoing evaluation process.

Two further structural questions also require decisions. The first is authority. Organizations have to decide whether the committee's recommendations are advisory or binding and what happens when its analysis conflicts with a decision leadership has already made (e.g., a tool purchased for its expected billing efficiency). An advisory committee whose recommendations can be overridden is unlikely to have the intended impact. At minimum, AI committee analyses should be documented, leadership should respond in writing when it chooses not to follow a recommendation, and tools used in clinical decision-making should not be deployed without AI committee review.

The second is conflict of interest. Committee members may have relationships with vendors (e.g., consulting, equity, or referral arrangements), and organizations may have already invested in a tool before the committee reviews it. Both create contingencies favoring adoption regardless of the analysis. Members should disclose vendor relationships and recuse themselves from related decisions. And, the committee should state explicitly when it is reviewing a tool the organization has already purchased.

Scaling AI Governance to Organizational Size

The committee structure described above assumes an organization large enough to staff it. Many behavior analysts may work in solo practices or small agencies where a standing committee with technical expertise is not feasible. These settings might be at greater risk because it is difficult to develop any formal review process. However, the analytic

process does not require a committee, *per se*. Rather, it requires that someone work through the four steps before a tool is adopted and revisit the steps as the tool changes. How that happens can scale with organizational size.

For solo practitioners, a minimum viable version of the process might involve answering, in writing, a short set of questions before adopting a tool. What does the tool do with client data, and are those data used to train the vendor's models? Given how the tool works, what kinds of errors is it likely to make? How will I verify its outputs before they influence a clinical decision or a signed document? What will I tell clients and caregivers? And, when will I re-evaluate? Periodic consultation with a trusted colleague outside one's practice can also serve as a proxy for the multiple perspectives a committee provides.

Small agencies might also share a committee. Several practices could pool clinical and administrative representatives, contract jointly for technical consultation, and share their tool evaluations with one another. Professional associations and state chapters are also well positioned to offer review resources such as published evaluations of commonly marketed tools, model consent language, and vendor questionnaires. Mid-size organizations might fold AI review into an existing ethics committee (Cox, 2020) with a designated member responsible for AI and an external technical consultant available as needed. The full committee described above is likely most realistic for larger organizations and for any organization adopting higher-risk tools (Table 2). Across all sizes, the depth of review should be proportional to the risk of the tool (Cox, 2025). For example, a drafting assistant for nonclinical emails warrants far less scrutiny than an agentic system that recommends treatment changes.

Activities of AI Committees

The specific activities of an AI committee reflect the ethical challenges described earlier in this article as well as those unaddressed (Table 3) and those that arise in the future. For example, hallucination, automation bias, bias in training data, and the opacity of model behavior influence practical decisions about whether a tool should be adopted, how it should be used, and what safeguards must be implemented. AI committees translate these concerns into organizational practice. In doing so, the committee tailors the analytic approach demonstrated in the preceding sections. Rather than asking whether AI is broadly ethical, the committee repeatedly asks more practical questions: What specific risks does this specific tool create for the populations we serve, and how should the organization respond? This approach mirrors recommendations from the broader AI governance literature that emphasizes context-specific evaluation of AI systems rather than universal ethical rules (Floridi et al., 2018; Jobin et al., 2019). Table 5 describes the core functions that AI committees might consider adopting.

A Worked Example: Evaluating a Longitudinal Clinical Co-Pilot

To make the analytic method and AI committee structure and function described in this article concrete, this final section works through a putative, emerging AI system in ABA: the LLM-driven longitudinal clinical co-pilot. We chose this system deliberately as a high-risk stress test. Persistent memory and autonomous operation magnify nearly every concern discussed above. The same process applies to lower-risk tools as described at the end of this section. The example below illustrates how an organizational AI committee might apply the analytic method outlined earlier in this article. That is, understanding the system's functional properties, identifying the ethical risks those properties create, analyzing those risks through multiple ethical frameworks, and translating that analysis into governance decisions.

How might an LLM-based longitudinal clinical co-pilot work in this worked example? Perhaps, after each session, the RBT uploads session data, notes, and video recordings of the session in a multimodal implementation. Between sessions, autonomous AI agents operate continuously. They maintain persistent memory for each client, draft weekly trend analyses, predict mastery timelines, and alert the supervising BCBA to regression or plateau patterns. The BCBA reviews these outputs through a dashboard and accepts, edits, or overrides the agent's recommendations. Six months into treatment, the agent has effectively ingested the entire clinical record associated with the client.

An AI committee must determine whether the organization should adopt this tool and, if so, under what conditions. Following the analytic method modeled earlier in this article (Figure 2), the committee begins by first examining how the system functions; then identifying the ethical challenges created by that functionality; third, analyzing those challenges through the converging lenses of behavior analytic ethics, clinical ethics, and AI ethics; and finally recommending conditions to adopt and own AI system performance monitoring if it is adopted.

Step 1: Understanding the Functional Properties of the System. The committee's first task is understanding how the tool behaves at a functional level. Unlike documentation tools that generate isolated outputs, the defining property of a longitudinal clinical co-pilot is persistent memory. Each output generated by the system is influenced by the entire history of previously ingested and created records. In effect, the system constructs an evolving narrative model of the client's behavioral history as digitally described. From an ethical standpoint, the critical property is that the AI system

retains and continuously reinterprets past information which then influences generated recommendations. In other words, the AI system constructs a cumulative interpretation of the client's clinical trajectory in addition to producing notes and suggestions.

Step 2: Identifying the Primary Ethical Risks. Once the system's functional architecture is understood, the committee identifies the risks that emerge from those properties. Many ethical concerns are possible, but one risk quickly becomes central: how do errors behave in a system that remembers. A session-note generator produces errors bounded to a single document. In contrast, an agent with persistent memory produces errors that compound over time. A small misclassification in the second week of therapy (e.g., incorrectly labeling a prompt level or behavior category) becomes a baseline assumption embedded within the agent's digital representation of the client by week 24. The harm, therefore, is not a single inaccurate recommendation. Rather, it is the gradual creation of a digital twin[5] for the client whose description of the child's behavior history may slowly diverge from reality while still appearing coherent and authoritative. This dynamic illustrates a broader feature of agentic systems. Agentic system outputs are the distributional result of hundreds or thousands of micro-decisions about what information to attend to and how that information is weighted, summarized, stored, retrieved, and discarded. Thus, there is rarely a single output that can be evaluated in isolation.

Step 3: Analysis Through Converging Ethical Frameworks. The committee then analyzes the identified risks through the same ethical frameworks discussed earlier in the manuscript. From a behavior analytic ethics perspective, the BACB Ethics Code (2020) assumes that behavior analysts base clinical decisions on data and professional judgment. In traditional supervision models, a BCBA reviewing a case has direct access to the raw datasets and the reasoning behind clinical decisions. A longitudinal AI co-pilot changes that dynamic. When recommendations are generated through months of autonomous processing across large datasets, the reasoning pathway that produced a recommendation may no longer be fully inspectable by the practitioner responsible for the case. The supervising BCBA may be asked to approve or reject recommendations derived from a narrative model that neither the practitioner nor the committee can fully reconstruct nor fully course-correct.

From a clinical ethics perspective, the tool implicates multiple principles simultaneously. Beneficence and non-maleficence are implicated through the possibility that accumulating errors influence treatment decisions. Autonomy is implicated because caregivers may not understand how their child's data are being interpreted by a continuously operating algorithmic system. Justice arises through the possibility that errors compound differently across clients depending on variations in documentation practices, treatment fidelity, or data quality.

From an AI ethics perspective, the tool introduces challenges related to explicability and accountability. Persistent memory systems blur the boundary between discrete outputs and evolving digital representations. If an algorithm's interpretation of a client's behavioral trajectory changes over time due to model updates or internal compression of stored information, the practitioner and organization may be unable to determine which underlying factors produced a recommendation.

Step 4: Translating the Analysis into Governance Decisions. The committee completes this step in two parts: gathering the information it needs from the vendor (Step 4a) and setting the conditions of adoption (Step 4b).

Step 4a: Committee Deliberation and Vendor Inquiry. With these concerns identified, the committee shifts to evaluating the vendor's system design and governance practices. Rather than asking only whether the tool performs well in demonstrations, the committee asks questions that follow directly from the ethical analysis: What information is written to the agent's persistent memory and what is discarded? Can that memory be inspected or audited by the organization? How are digital representations corrected when errors are found? When the vendor updates the underlying model, how does the system handle previously stored client information? Does the client's longitudinal representation transfer intact, become compressed, or regenerate anew under the updated model? Where is the client's data stored, and does persistent memory distribute protected health information across multiple model providers or cloud environments? Are client data used to train or improve the vendor's models, and can the organization opt out? These questions reflect a shift in how AI tools must be evaluated. The ethical challenge is no longer limited to assessing individual outputs. Instead, the committee is tasked with evaluating how the system constructs and maintains its internal representation of the client over time.

Step 4b: Conditional Adoption and Governance Safeguards. Given the dynamic nature of such systems, the committee's decision is not a binary approve-or-reject determination. Instead, adoption becomes contingent on governance safeguards designed to preserve clinical oversight. The committee therefore proposes several conditions

[5]A digital twin is a virtual representation of a physical asset via collected data (e.g., Emmert-Streib, 2023). Often, digital twins are combined with simulation to predict, optimize, monitor, control, and improve decision-making relative to that physical asset. In the current context, a digital twin would refer to the total clinical picture available to any AI system defined solely by the data entered and available to that AI system.

for adoption. First, the use of persistent AI memory requires periodic reconsent rather than a single consent at intake. Families must be informed that their child's data are being incorporated into a continuously evolving system. Second, supervising BCBAs must conduct periodic clean-slate reviews. That is, at predetermined intervals, the clinician reviews the client's raw data independently of the agent's summaries before accepting any recommendations. And, the ABA organization rearranges their operational contingencies to allow for these periodic intensive reviews. Third, AI outputs cannot serve as the sole basis for clinical decisions such as goal modification, mastery determinations, or discharge planning. Fourth, the AI committee conducts quarterly audits comparing the agent's internal summaries with the underlying clinical record. Behavior analysts already have a well-developed technology for this kind of comparison in interobserver agreement (IOA; Cooper et al., 2020). The audit can, therefore, be structured as an IOA procedure. An independent clinician reviews the raw data and scores the agent's summary for agreement across operationally defined categories (e.g., target behaviors, prompt levels, mastery status, reported trends). Finally, any divergence between the system's representation of the client and the underlying data that exceeds a threshold set in advance by the organization (e.g., agreement below 80% on any category) triggers a formal review of continued tool use. And, rollback processes are in place if immediate removal of a tool is found warranted.

Implications for AI Governance in ABA. Though incredibly brief, this example illustrates a broader point. AI ethics in ABA relative to LLMs, multimodal systems, and generative AI is unlikely to be governed primarily by one-time decisions about whether a technology is ethical. Instead, it will increasingly involve ongoing governance processes in which organizations continuously monitor how these probabilistic systems interact with clinical practice. In this sense, the function of an AI committee is to both approve technologies and to create the institutional infrastructure necessary for responsible experimentation and ongoing monitoring of new tools. The adoption decision becomes the beginning of an ongoing process of observation, measurement, and adjustment rather than the end of ethical deliberation.

The same process applies to lower-risk tools and would likely proceed faster. Consider an enterprise LLM that drafts routine scheduling messages to caregivers from a staff member's bullet points. Its functional properties are simple. A staff member supplies the content and the tool generates prose without accessing clinical data or retaining information across uses (Step 1). The primary risks are minor factual errors and a tone that may not match the family's preferences (Step 2). Viewed through the frameworks above, the benefits (e.g., time saved) are real and the potential harms are small and easily caught (Step 3). A reasonable governance decision is approval with a simple safeguard that staff read and attest to the accuracy of every message before sending it (Step 4). Note that proportionality is the point. The analytic process stays the same, but the depth of analysis and the safeguards it produces should scale with the risk of the tool (Cox, 2025).

Summary

The field of applied behavior analysis finds itself in a position familiar to many industries. A new technology is being widely adopted before the ethical frameworks for evaluating it have fully caught up. This is not a crisis but, rather, the normal condition of practice in a rapidly changing technological environment. But it does demand a response. Existing ethical guidance on AI in ABA currently comes from four primary sources (AIC-ABA, 2025; CASP, 2025; Cox, 2025; Jennings & Cox, 2024). Together these works provide an important foundation for thinking about AI ethics in the field. However, each addressed AI broadly. This article narrowed the analytic lens to LLMs and LLM-supported tools, which likely dominate most practitioners' day-to-day interactions with AI systems.

Prior work on AI ethics in ABA has largely taken an operational orientation. Here high-level guidance on how practitioners and organizations might respond when AI-related issues arise has been offered. The contribution of the present article is to extend past conversations to add an analytical layer to examine why specific mechanisms of LLM-based systems create distinct ethical risks and how practitioners might reason through those mechanisms when evaluating emerging tools. Four challenges associated with LLM-based tools were examined in depth: hallucination and its interaction with clinical de-skilling and never-skilling; bias embedded in internet-scale training data; the opacity of LLM systems and its implications for informed consent; and the risks associated with AI-generated clinical documentation. Each challenge was analyzed through converging perspectives drawn from clinical ethics, behavior analytic ethics, and AI ethics. A common thread emerged across these challenges. AI tools can leave the products of clinical work intact while removing the behavior that made those products clinically meaningful.

More broadly, this article offers a potential general analytic approach that practitioners and organizations can use when evaluating emerging AI tools. Specifically, it comprises four steps. First, understand the system's functional mechanisms. Second, identify the ethical risks those mechanisms create. Third, analyze those risks through converging ethical frameworks. And, lastly, translate that analysis into governance decisions for responsible use. Such a technology-specific ethical analysis should become part of the routine operating practice of organizational AI committees. To

support translation from theory to practice, we presented a worked example illustrating how such a committee might evaluate an emerging AI tool, including the novel governance questions raised by longitudinal and agentic AI systems.

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Tables

Table 1. Existing ethical guidance on AI in ABA.

Document	Year	Type	Scope	Key Contributions	LLM-Specific?
Starting the Conversation Around the Ethical Use of AI in ABA (Jennings & Cox, 2024)	2024	Peer-Reviewed Article	AI Broadly	First known AI ethics article in ABA; principlist framework; integrated clinical ethics in ABA with AI ethics	No
Ethical Behavior Analysis in the Age of AI (Cox, 2025)	2025	Peer-Reviewed Article	AI Broadly	AI literacy as a prerequisite for ethical AI use; understanding model-building; education and training recommendations	No
Practice Parameters for AI Use in ABA (CASP, 2025)	2025	Organizational Practice Parameters	AI Broadly	Organizational AI Lifecycle (selection, deployment, monitoring, auditing, deprecation); Legal/regulatory overview; Glossary of AI terms	No
Ethical Guidelines for AI in ABA (AIC-ABA, 2025)	2025	Organizational Ethical Guidelines	AI Broadly	11 ethical principles for AI in ABA; Broad aspirational standards for the field	No

Note. All four documents provide valuable guidance for the ethical use of AI in ABA. Each was appropriately scoped to address AI broadly given the state of the conversation at the time of writing.

Table 2. Types of LLM-based systems behavior analysts are likely to encounter.

System Type	Example Uses	Typical Safeguards	Relative Risk for Clinical Use
General-Purpose Public LLM	A practitioner enters a clinical question or client details into a consumer chatbot.	Few. Entered data may be retained or used for model training unless user opts out.	Highest. Generally inappropriate for client data or clinical decisions.
Enterprise or Private LLM Deployment	An organization-licensed assistant used for drafting, summarizing, or brainstorming.	Contractual data protections, administrator controls, and usage logs. Outputs not grounded in verified sources.	High for clinical or educational. Lower for non-clinical or non-educational.
Retrieval-Augmented Tool	Questions answered from an organization's policies, published guidelines / articles, client's own records, etc.	Outputs grounded in curated sources; often with citations to those sources.	Moderate. Depends on the quality of the sources and on users checking what is cited.
ABA-Specific Documentation Product	Session notes, progress reports, or treatment plan drafts generated from session data.	Templates and structured fields tied to session data; BCBA review and signature.	Moderate to high. Few published validation studies in ABA or behavioral health.
Agentic or Multimodal System with Persistent Memory	A longitudinal co-pilot that ingests session data and video. Drafts trend analyses and goal or target recommendations.	Varies widely. Memory auditing and limits on autonomy are rarely standard.	Highest. Errors can compound across months of outputs and unknown or unmonitored data sharing.

Note. Risk levels are relative and illustrative. The risk of any specific tool depends on its design, the data it accesses, the decisions it informs, and the safeguards in the surrounding workflow. Documentation-related functions also vary in risk, from drafting and summarizing (lower) to interpreting, recommending, and autonomous decision support (higher).

Table 3. Ethical challenges posed by LLMs and LLM-driven tools in ABA.

Challenge	Description	Unique or Heightened by LLMs?	Analyzed Here?	Where Else Addressed or Why Deferred
Hallucination / Factual Unreliability	Generation of confident, fluent, factually incorrect content (e.g., fake citations).	Unique to generative AI; Structural feature of LLMs	Yes	Glossary and staff training guidance in CASP (2025).
Automation Bias and De-Skilling or Never-Skilling	Overreliance on AI outputs; reduction in unprompted performance level or failure to develop independent skills.	*Heightened.* LLM outputs are often uniquely fluent and read authoritative.	Yes	*Not previously addressed in ABA and AI literature.*
Bias in Training Data and Systems	Internet-scale training data encodes societal biases reproduced in outputs.	*Heightened.* Training data is significantly larger and less characterizable.	Yes	Addressed at a general level in prior work (Table 1).
Informed Consent and Opacity	Inability to explain how AI system generates its outputs. Creates a layered consent challenge.	*Heightened.* LLMs are significantly more opaque than previous model-driven systems.	Yes	Jennings and Cox (2024); CASP (2025); AIC-ABA (2025).
Clinical Documentation Generation	Risks of inaccuracy, bias, and de-skilling in AI-generated session notes and treatment plans.	Unique to generative AI as documentation tool.	Yes	*Not previously addressed in ABA and AI literature.*
Data Privacy & PHI in AI Systems	Clinical data transmitted to third-party servers; consumer vs. enterprise distinction; HIPAA compliance.	*Heightened.* Data sent to external APIs rather than processed locally.	Partly	Data transmission & HIPAA compliance addressed in CASP (2025)
Accountability and Liability	Unclear responsibility when AI contributes to decision-making errors; emerging legal frameworks.	*Heightened.* LLM outputs are harder to audit and attribute.	No	Warrants dedicated analysis.
Agentic AI and Compound Error	Autonomous, multi-step AI systems that can cascade errors across clinical or educational workflows.	Unique to agentic AI systems.	Partly	Analyzed in the worked example. Warrants dedicated analysis
AI Companions / Tutors & Vulnerable Populations	Parasocial relationships between AI chatbots and vulnerable individuals (e.g., minors with ASD or DD).	Unique to generative AI interaction design.	No	Warrants dedicated analysis.
AI in Prior Authorization	AI-driven insurance denial systems affecting access to ABA services.	Not LLM-specific, but increasingly AI-driven.	No	Warrants dedicated analysis.

Note. This table is not exhaustive. Challenges were included based on their presence in the clinical ethics, behavior analytic ethics, or AI ethics literatures through March 2026 when the manuscript was submitted. The four challenges analyzed in depth in this article were selected for being clearly LLM-specific or LLM-heightened, clinically concrete, analytically rich, and collectively representative of different types of ethical risk.

Table 4. A practitioner reference for a current ethical analysis of LLM challenges in ABA.

Challenge	Core Clinical Concern	Key AI Ethics Principles	BACB Code (2020) Standards	Questions for Practitioners and AI Committees
Hallucination and De-skilling	AI system generates plausible but incorrect clinical content. Practitioners may not catch errors. Skills may reduce or never develop.	Explicability; Non-maleficence; Ethics of Care	Practice within Scope of Competence; Maintaining Competence; Providing Effective Treatment	Can staff identify hallucinated content reliably? What verification processes are in place? Are new practitioners developing independent clinical skills or relying on AI from the start?
Bias in Training Data	Internet-scale biases reproduced in clinical outputs; stigmatizing language; worse performance for underrepresented populations.	Justice; Ethics of Care	Cultural Responsiveness and Diversity; Nondiscrimination; Awareness of Personal Biases and Challenges	Has the tool been evaluated for performance across the populations we serve? Who monitors for biased language in AI-generated content? How do we detect subtle stigmatizing patterns?
Informed Consent / Opacity	Neither practitioner nor developer can fully explain LLM outputs. Clients cannot meaningfully evaluate AI's role.	Explicability; Autonomy; Ethics of Care	Communicating About Services; Involving Clients and Stakeholders; Obtaining Informed Consent	Do consent forms disclose AI use? Can staff explain what the tool does and cannot do? Do clients know they can opt out? Is consent revisited when tools change?
Clinical or Educational Documentation	Inaccurate technical language; subtle bias in framing; de-skilling of analytical documentation process.	Beneficence; Explicability	Being Truthful; Documentation Protection and Retention; Accuracy in Service Billing and Reporting; Collecting and Using Data	Are AI-generated notes verified against raw data before signing? Does AI use in documentation change how thoroughly staff engage with client progress? Are clients informed?

Note. This table is intended as a practical reference for behavior analysts and organizational AI committees evaluating LLM-based tools. BACB Ethics Code references are to the *Ethics Code for Behavior Analysts* (BACB, 2020). The questions listed are illustrative, not exhaustive.

Table 5. Core functions AI committees might adopt.

Function	Description
Prospective Ethical Analysis of AI Tools	Before the organization adopts a new AI tool, the committee conducts an analysis like the one modeled. The analysis walks through clinical ethics, behavior analytic ethics, and AI ethics lenses to identify the specific risks the tool creates for the specific populations served.
Ongoing Monitoring and Review of AI Tool Performance	Because AI systems change over time, a tool that was ethically acceptable at adoption may not remain so. Committees should review AI outputs across served populations, track reported errors, and reassess considering new research or vendor updates. CASP's (2025) monitoring and auditing guidelines provide a useful operational framework.
Policy Development	Employees using AI tools need clear policies on which tools are approved for what purposes, what data may be input to AI systems, what verification is required before AI-generated content is used clinically, what consent disclosures are required, and how AI-related errors should be reported.
Staff Education	All staff using AI tools should receive training commensurate with their role, including how the tools work, what their limitations are, and what organizational policies require. Cox (2025, Table 2) outlines AI literacy levels by behavior analytic role.
Incident Review and Learning	When AI-related errors or ethical concerns arise, the committee reviews them to identify systemic issues and improve practices, not to assign blame.
Vendor and Contract Review	Many safeguards are contractual rather than clinical. Before adoption and at each renewal, the committee reviews vendor terms covering data retention and deletion, whether client data may be used to train or improve the vendor's models, notification of model updates, the organization's access to logs for auditing, rollback and exit procedures, and HIPAA business associate agreements.

Figures

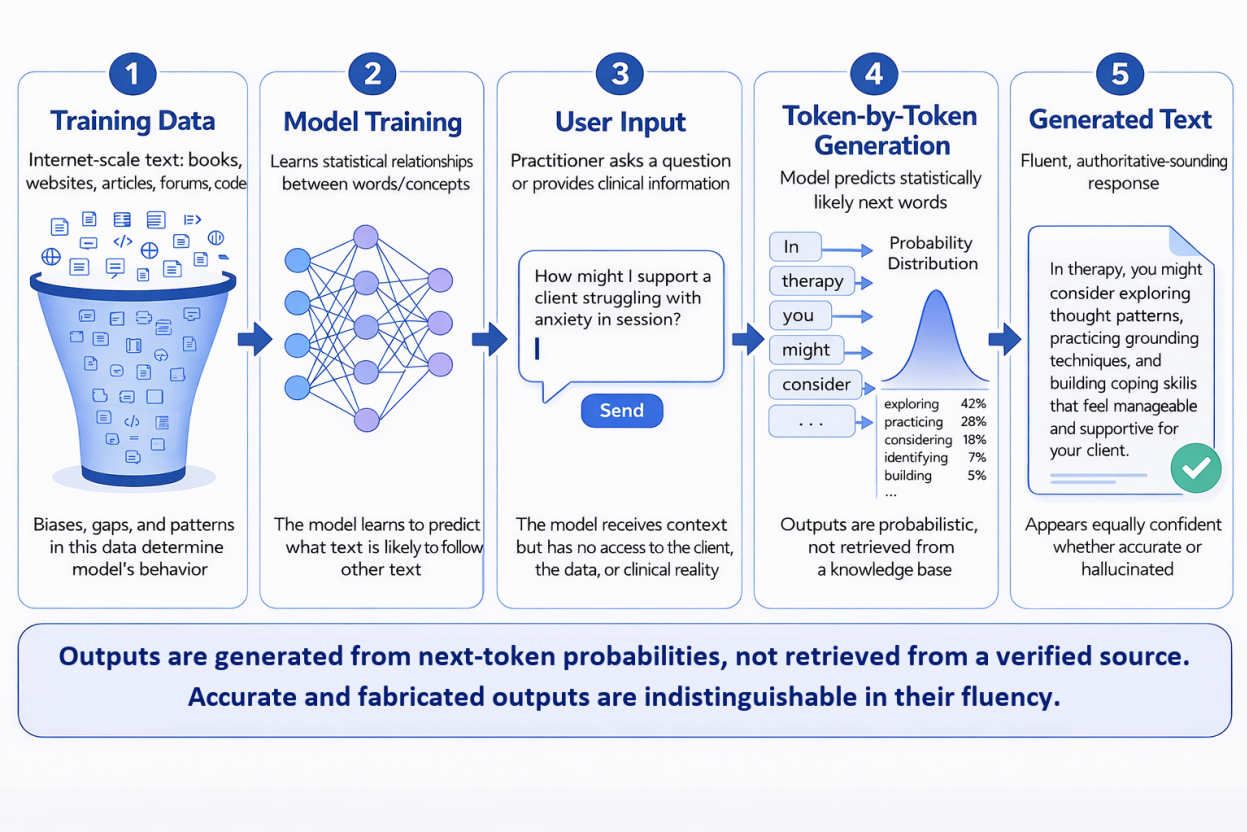


Figure 1. Overview of how LLMs generate output.

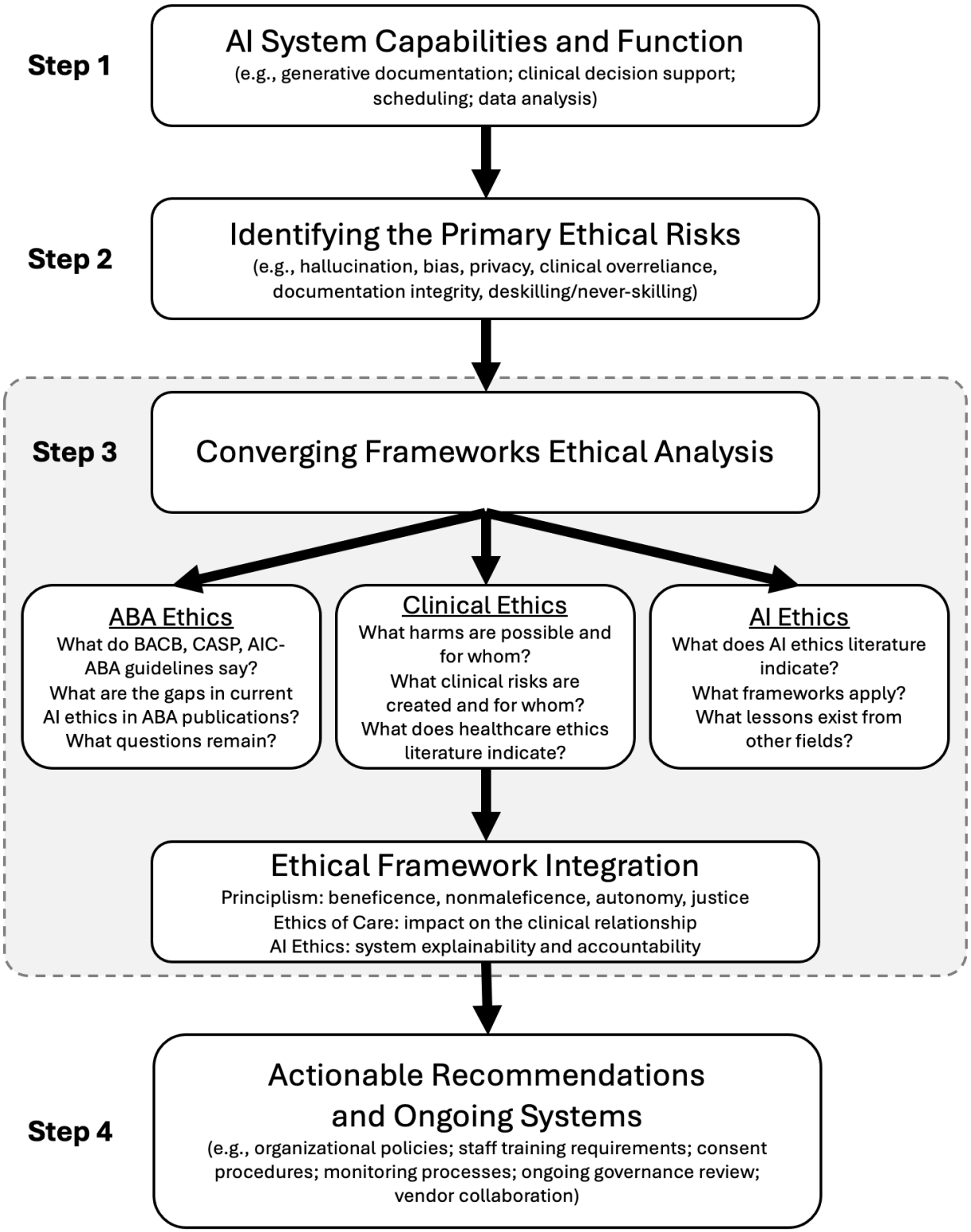


Figure 2. Example multi-framework ethical analysis sequence for AI tools in ABA.